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Biological object image-capturing and treatment system and method

Abstract

A biological object image-capturing and treatment system includes a micro detection and treatment device, and an AI model. The micro detection and treatment device includes a plurality of signal transmitting and receiving elements, wherein one of the signal transmitting and receiving elements continuously transmits a stimulation signal with different parameter values to a particular target, and one of the signal transmitting and receiving elements continuously receives response signal from the particular target. The AI model uses the response signals to perform machine learning, and outputs a parameter value adjustment suggestion to the micro detection and treatment device according to result of the machine learning. The micro detection and treatment device adjusts the parameter value of the stimulation signal according to the parameter value adjustment suggestion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefits of the Taiwanese Patent Application Serial Number 108133280, filed on Sep. 16, 2019, the subject matter of which is incorporated herein by reference. (2) This application is a continuation in part (CIP) of U.S. patent application for “Biological object image-capturing and treatment system and method”, U.S. application Ser. No. 16/902,300 filed Jun. 16, 2020, and the subject matter of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a biological object image-capturing and treatment system with low noise and a method thereof.

2. Description of Related Art

(2) Radiological medicine technology has been widely used in medical treatment behavior, such as ultrasound image-capturing technology or ultrasound therapy technology. Ultrasonic technology is provided with the following advantages: (1) it is less radioactive and therefore less harmful to the human body in comparison with other technologies; (2) its price is lower than other inspection techniques; (3) its equipment volume is smaller, and thus more flexible; and (4) the image-capturing speed is fast and the resolution is high. However, ultrasonic probes typically use piezoelectric transducers as signal transmitting and receiving components, and the materials of these transducers include piezoelectric polycrystalline ceramic materials, wherein the silver surface plated on the ceramic materials is connected to the electrode of the next transducer. Therefore, these ceramic materials are likely to produce resonance and, when adjacent transducers transmit or receive signals, the problem of signal interference often occurs.

(3) In addition, there is currently no ultrasonic device into which the image-capturing function and the treatment function are integrated, and thus there are still deficiencies in terms of space occupation, production cost, or usage convenience.

(4) In addition, different biological objects or different positions may have different absorption degrees for ultrasonic signal, so that how to select ultrasonic signal with an appropriate power level for different biological objects or different positions becomes a problem that needs to be solved. So far, this problem can be solved only by manual adjustment, and the efficiency is very low.

(5) Therefore, it is desirable to provide an improved biological object image-capturing and treatment system and method to mitigate and/or obviate the aforementioned problems.

SUMMARY OF THE INVENTION

(6) The object of the present invention is to provide a biological object image-capturing and treatment system and method, with which not only the image-capturing function, the treatment function and the stimulation function are integrated into the same device for reducing the occupied space of the device, but also special signal transmitting operation modes are employed to greatly reduce the interference between transmitting and receiving elements of each signal in the device, thereby significantly improving the signal accuracy. Besides, the biological object image-capturing and treatment system is integrated with an artificial intelligence model for automatically adjusting the parameter values of the ultrasonic signal transmitted from the biological object image-capturing and treatment system.

(7) In one aspect of the present invention, there is provided a biological object image-capturing and treatment system for performing an object stimulation procedure. The system comprises a micro detection and treatment device including a plurality of signal transmitting and receiving elements and an AI model. One of the signal transmitting and receiving elements continuously transmits a stimulation signal with different parameter values to a particular target, and one of the signal transmitting and receiving elements continuously receives response signal corresponding to the stimulation signal with different parameter values from the particular target. The AI model uses the response signals corresponding to the stimulation signal with different parameter values from the particular target to perform a machine learning, and outputs a parameter value adjustment suggestion to the micro detection and treatment device according to result of the machine learning. The micro detection and treatment device adjusts the parameter value of the stimulation signal according to the parameter value adjustment suggestion.

(8) In another aspect of the present invention, there is provided a biological object image-capturing and treatment method performed by a biological object image-capturing and treatment system. The

biological object image-capturing and treatment system comprises a micro detection and treatment device including a plurality of signal transmitting and receiving elements and an AI model. The method includes the steps of: continuously transmitting a stimulation signal with different parameter values to a particular target by one of the signal transmitting and receiving elements; continuously receiving response signal corresponding to the stimulation signal with different parameter values from the particular target by one of the signal transmitting and receiving elements; using the response signals corresponding to the stimulation signal with different parameter values to perform a machine learning by the AI model, and outputting a parameter value adjustment suggestion according to result of the machine learning to the micro detection and treatment device by the AI model; and adjusting the parameter value of the stimulation signal according to the parameter value adjustment suggestion by the micro detection and treatment device.

(9) Other objects, advantages, and novel features of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating the biological object image-capturing and treatment system according to an embodiment of the present invention;
- (2) FIG. 2 schematically illustrates a controller of a micro detection and treatment device according to an embodiment of the present invention;
- (3) FIG. 3(A) schematically illustrates a signal transmitting and receiving element array according to an embodiment of the present invention;
- (4) FIG. 3(B) schematically illustrates a user operating device corresponding to the signal transmitting and receiving element array according to an embodiment of the present invention;
- (5) FIG. 4 is a circuit diagram of a mode control module and a startup module according to an embodiment of the invention.
- (6) FIG. 5 is a schematic diagram illustrating the biological object image-capturing and treatment system according to another embodiment of the present invention; and
- (7) FIG. 6 is a flow chart illustrating the steps of the object stimulation procedure of the biological object image-capturing and treatment method according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (8) The implementations of the present disclosure will be described with specific embodiments in the following description. A person skilled in the art will understand the advantages and the effects provided by the present disclosure. Different specific embodiments may be applicable according to the present disclosure.
- (9) Moreover, the orders such as “first”, “second”, and so on, in the specification and the claims are only used to distinguish the elements with the same name. They do not have their own specific meanings, do not necessarily mean that there is another element existing in addition to one element, and do not mean that there is a priority between one element and another element, or one step and another step.
- (10) Besides, in the present disclosure, the description such as “A happens when B happens” may refer to A happens before, when or after B happens, and it does not necessarily refer to A and B happen at the same time, except that a clear limitation is given. In the present disclosure, the description such as “A is disposed on B” refers to the corresponding locations of A and B, and it does not necessarily refer to the contact of A and B, except that a clear limitation is given. Moreover, the word “or” between the elements or their effects in the present disclosure means that the elements and the effects may exist individually or together.

(11) In addition, in the present invention, the terms such as “connected”, “electrically connected” or “coupled”, unless otherwise emphasized, mean that the direct connection and the indirect connection are included. In addition, the terms “including”, “including”, “having”, and “having” in this disclosure are all open descriptions, which are described first.

(12) Furthermore, various embodiments of the biological object image-capturing and treatment system **1** of the present invention can be realized by a software program or an electronic circuit, and are not limited thereto.

(13) FIG. **1** is a schematic diagram illustrating the biological object image-capturing and treatment system according to an embodiment of the present invention, wherein the biological object image-capturing and treatment system **1** may be used to perform an image-capturing procedure, a treatment procedure or an object stimulation procedure. As shown in FIG. **1**, the biological object image-capturing and treatment system **1** at least includes one micro detection and treatment device **10**. Further, the biological object image-capturing and treatment system **1** may further include a user operating device **150** and a cloud server **250**. The micro detection and treatment device **10** includes a signal transmitting and receiving element array **12**, and the signal transmitting and receiving element array **12** includes a plurality of signal transmitting and receiving elements **120**, wherein the plurality of signal transmitting and receiving elements **120** can be arranged in an array, but it is not limited thereto. The signal transmitting and receiving elements **120** are configured in such a manner that: the adjacent at least two signal transmitting and receiving elements **120** respectively transmit signals or receive signals during different periods; that is, the period during which one signal transmitting and receiving element **120** transmits or receives signals does not overlap the period during which another signal transmitting and receiving element **120** transmits or receives signals. It should be noted that the “period” herein may be in a state of continuous time or instantaneous time. In addition, the micro detection and treatment device **10** may also include a controller **16** for controlling the operation of the micro detection and treatment device **10**. In one embodiment, the micro detection and treatment device **10**, the user operating device **150** and the cloud server **250** may transmit data to each other. The image-capturing procedure and the treatment procedure may be handled by separate subsystems, or the image-capturing procedure and the treatment procedure may be handled by a common system.

(14) The micro detection and treatment device **10** is a medical detecting device, and has both an image-capturing function and a treatment function, wherein “micro” is defined as a size capable of being inserted into a human body. However, the micro detection and treatment device **10** can also be used outside the human body. In one preferred embodiment, the micro detection and treatment device **10** is an ultrasonic detection and treatment device, and thus the signal transmitting and receiving element **120** is used for transmitting and receiving ultrasonic signals, but is not limited thereto. For convenience of description, the following description is given by taking the transmitting and receiving elements to transmit or receive ultrasonic signals as an example. When the biological object image-capturing and treatment system **1** performs an image-capturing process, the micro detection and treatment device **10** may transmit a first power signal to different positions of a target object **500** through the signal transmitting and receiving elements **120**, and may receive signals reflected from different positions of the target object **500** through the signal transmitting and receiving elements **120**. Furthermore, the reflected signals may be integrated through the user operating device **150** or the cloud server **250** so as to combine an ultrasonic image of the target object **500**. When the biological object image-capturing and treatment system **1** performs a treatment procedure, the micro detection and treatment device **10** may transmit a second power signal to a specific position of the target object **500** through the signal transmitting and receiving elements **120**, thereby treating the specific position of the target object **500**, for example, ablation and other treatment behaviors. It should be noted that the first power signal and the second power signal have different powers or different frequencies. In one embodiment, the power of the first power signal is less than the power of the second power signal. In one embodiment, the frequency

of the first power signal is higher than the frequency of the second power signal. As a result, the first power signal is suitable for object image-capturing, and the second power signal is suitable for treatment behavior such as ablation. Besides, when the object stimulation procedure is performed, the signal transmitting and receiving elements **120** can transmit stimulation signal to an object, wherein each of the signal transmitting and receiving elements **120** corresponds to different object or different position of the same object. The stimulation signal can be ultrasonic signal. In one embodiment, the power of the stimulation signal is smaller than that of the first power signal and the second power signal, but it is not limited thereto.

(15) In addition, the micro detection and treatment device **10** may be, for example, an invasive medical device, so that it has a smaller volume than the existing handheld ultrasonic device. In FIG. **1A**, the target object **500** is exemplified by the heart of a human body, and the micro detection and treatment device **10** may be placed near the heart to perform image-capturing and treatment on the heart vascular wall, vein, and the like. In one embodiment, the micro detection and treatment device **10** may be of medical grade material suitable for placement into a human body. In one embodiment, at least a portion of the micro detection and treatment device **10** is a disposable component and can be replaced.

(16) In one embodiment, the diameter of each of the signal transmitting and receiving elements **120** is equal to or smaller than 10000 micrometers (μm). In one embodiment, the diameter of each of the signal transmitting and receiving elements **120** is in a range of 4000 μm to 6000 μm (i.e. $4000\ \mu\text{m} \leq \text{diameter} \leq 6000\ \mu\text{m}$). In one embodiment, the diameter of each of the signal transmitting and receiving elements **120** is substantially 5000 μm . However, the invention is not limited thereto.

(17) Since the micro detection and treatment device **10** of the invention can be inserted into the human body, compared to the prior technique, the micro detection and treatment device **10** is closer to a treatment target. Furthermore, because of having small volume, the signal transmitting and receiving elements **120** can provide more precise and accurate imaging distance, imaging depth and imaging width, so as to make a reflection of dose-response relationship more accurate.

(18) Besides, since the micro detection and treatment device **10** may perform an image-capturing procedure and a treatment procedure, the present invention may further perform steps to achieve treatment and resection of the proliferative tissue at a specific position in the human body.

(19) Please refer to FIG. **1** again. The user operating device **150** may be connected to the micro detection and treatment device **10** in a manner of wired transmission or wireless transmission, whereby the user may transmit a control command to the micro detection and treatment device **10** through the user operating device **150**. In one embodiment, the user operating device **150** may be various electronic devices with a microprocessor, such as a desktop computer, a notebook computer, a smart phone, a tablet computer, a touch device, and the like, and is not limited thereto. More details regarding the user operating device **150** will be explained in more detail in the subsequent paragraphs with reference to FIG. **3(B)**.

(20) The cloud server **250** may perform data backup or data processing. In one embodiment, after the signal transmitting and receiving elements **120** receive the reflected signal from the target object **500**, instead of performing image formation, the micro detection and treatment device **10** transfers the data to the user operating device **150** or the cloud server **250** to perform image formation. In one embodiment, an AI model for performing machine learning can be disposed on the cloud server **250**, but it is not limited thereto.

(21) Next, the detailed structure of the controller **16** of the micro detection and treatment device **10** will be described.

(22) FIG. **2** schematically illustrates a controller **16** of a micro detection and treatment device **10** according to an embodiment of the present invention. With reference to both FIG. **1(A)** and FIG. **2**, the controller **16** is disposed inside the micro detection and treatment device **10**. The controller **16** may be coupled to the signal transmitting and receiving element array **12** to control the operation of the signal transmitting and receiving elements **120**. In one embodiment, the controller **16** may

include a main procedure selection module **20**, an address definition module **25**, a startup sequence setting module **30**, and a startup module **35**. Alternatively, the micro detection and treatment device **10** may further include a high voltage control module **40**, a delay module **50**, and a pulse modulation module **60**. The type and the quantity of the aforementioned modules are only examples and are not limited, and may be arbitrarily increased or decreased according to the needs of the user. In addition, the modules may be arbitrarily connected or integrated as long as they can be realized. In one embodiment, the controller **16** is a control chip, and the function of each module can be implemented by at least one digit logic circuit or computer program, and is not limited thereto.

(23) First, the main procedure selection module **20** is described. The main procedure selection module **20** may be used to execute an image-capturing procedure, a treatment procedure or an object stimulation procedure. For example, the main procedure selection module **20** selects the image-capturing procedure or the treatment procedure for execution according to a predetermined command or a control command of the user operating device **150**, and further controls the signal transmitting and receiving element array **12** to transmit the first power signal or the second power signal. In addition, in one embodiment, the main procedure selection module **20** may further select different image-capturing modes for execution. For example, the image-capturing procedure may include a first image-capturing mode, a second image-capturing mode, a third image-capturing mode and a fourth image-capturing mode, and the treatment procedure may include a first treatment mode and a second treatment mode, which will be described in order in the following.

(24) In one embodiment, the first image capturing mode is set to transmit the first power signal to different positions of the target object **500** by using a single signal transmitting and receiving element **120**, and receive reflection signals reflected from the different positions of the target object **500** at different time points by using a single signal transmitting and receiving element **120**. In this embodiment, the signal transmitting and receiving element that transmits the first power signal is not limited to be the same as the signal transmitting and receiving element that receives the reflection signals. In addition, since a single signal transmitting and receiving element **120** must transmit signals toward different positions of the target object **500**, the signal transmitting and receiving element array **12** is designed to be rotatable, but is not limited thereto.

(25) In one embodiment, the second image-capturing mode is set to transmit the first power signal to different positions of the target object **500** by using a plurality of signal transmitting and receiving elements **120**, and receive reflection signals reflected from the different positions of the target object **500** at different time points by using a single signal transmitting and receiving element **120**. In this embodiment, between the signal transmitting and receiving elements **120** that transmit signals, there is an interval of at least one signal transmitting and receiving element, and thus they are not adjacent to each other. In addition, signals may be transmitted simultaneously or sequentially. In addition, the signal transmitting and receiving elements **120** that transmit signals may also be used to receive signals.

(26) In one embodiment, the third image-capturing mode is set to transmit the first power signal to different positions of the target object **500** by using a single signal transmitting and receiving element **120**, and receive reflection signals reflected from the different positions of the target object **500** at different time points by using a plurality of signal transmitting and receiving elements **120**. Between the signal transmitting and receiving elements **120** that receive signals, there is an interval of at least one signal transmitting and receiving element, and thus they are not adjacent to each other. In addition, the signal transmitting and receiving element **120** that transmits signals may also be used to receive signals.

(27) In one embodiment, the fourth image-capturing mode is set to transmit the first power signal to different positions of the target object **500** by using a plurality of single signal transmitting and receiving elements **120**, and receive reflection signals reflected from the different positions of the target object **500** at different time points by using a plurality of signal transmitting and receiving

elements **120**. In this embodiment, the signal transmitting and receiving elements **120** that transmit signals are not adjacent to each other, and the signal transmitting and receiving elements **120** that receive signals are also not adjacent to each other. In this embodiment, the signal transmitting and receiving elements **120** that transmit signals may not be in consistency with the signal transmitting and receiving elements **120** that receive signals. In addition, signals can be transmitted simultaneously or sequentially.

(28) In one embodiment, the first treatment mode is set to transmit the second power signal to different positions of the target object **500** by using a single signal transmitting and receiving element **120**. It is noted that, when the first treatment mode is executed, the second power signal from the single signal transmitting and receiving element **120** does not need to be focused.

(29) In one embodiment, the second treatment mode is set to transmit the second power signal to the different positions of the target object **500** by using a plurality of signal transmitting and receiving elements **120**. In this embodiment, the signal transmitting and receiving elements **120** that transmit signals are not adjacent to each other. In addition, signals can be transmitted simultaneously or sequentially. It is noted that, when the second treatment mode is executed, the second power signal from the single signal transmitting and receiving elements **120** do not need to be focused.

(30) Next, the address definition module **25** is described. The address definition module **25** may define the address parameter of each signal transmitting and receiving element **120**. For example, if the signal transmitting and receiving element array **12** has 64 signal transmitting and receiving elements **120**, there are also **64** address parameters, and each signal transmitting and receiving element **120** corresponds to one address parameter.

(31) Next, the startup sequence setting module **30** is described. When using a plurality of signal transmitting and receiving elements **120** to transmit or receive signals, in order to avoid interference generated between the adjacent signal transmitting and receiving elements in operation, the startup sequence setting module **30** may control the signal transmitting and receiving elements **120** to perform a specific startup sequence mode so as to shift the transmitting and receiving timing of the adjacent signal transmitting and receiving elements **120**. The specific startup sequence mode is provided to sequentially or simultaneously start the signal transmitting and receiving elements in a manner of spacing N signal transmitting and receiving elements **120**, wherein N is a positive integer of 1 or more ($1 \leq N$), and N is smaller than the total number of the signal transmitting and receiving elements **120**.

(32) In one embodiment, the startup sequence setting module **30** may execute a first startup sequence mode, wherein the signal transmitting and receiving elements **120** are sequentially or simultaneously started in a manner of spacing one signal transmitting and receiving element **120**. Under this startup sequence, the signal transmitting and receiving elements of odd address parameter are started sequentially or simultaneously to receive or transmit signals. When the signal transmitting and receiving elements of odd address parameter complete the signal receiving or transmitting, the signal transmitting and receiving elements of even address parameter will start receiving or transmitting signals sequentially or simultaneously. In addition, it is also possible to first start the operation by using the signal transmitting and receiving element of even address parameter.

(33) On this basis, there N types of startup sequence modes are available for execution to prevent adjacent 1 to N signal transmitting and receiving elements **120** from performing signal transmitting or receiving during the same period, thereby avoiding interference problem among the ultrasonic signals. The N types of startup sequence modes may be suitable for signal transmitting or receiving in the image-capturing procedure or the treatment procedure.

(34) Next, the startup module **35** is described. The startup module **35** may generate a plurality of startup signals according to the startup sequence mode set by the startup sequence setting module **30**, wherein the startup signals correspond to the address parameters corresponding to the startup

sequence mode, and are used to actually start the signal transmitting and receiving elements **120** corresponding to the address parameters. For example, when the second startup sequence mode is set, the startup module **35** transmits the startup signal to the signal transmitting and receiving elements **120** corresponding to the address parameters according to the startup sequence of the second startup sequence mode (for example, first starting the address parameters of 1, 4, 7, 10 and so on, and then starting the address parameters of 2, 5, 8, 11 and so on), so as to start the signal transmitting and receiving elements **120** for performing signal transmitting or receiving.

(35) Next, the high voltage control module **40** is described. In one embodiment, since the voltage required for the signal transmitting and receiving elements **120** to transmit the ultrasonic signals is as high as 40 volts (40 V) or more, a high voltage must be applied to drive the signal transmitting and receiving elements **120** through the high voltage control module **40**, and thus the high voltage control module **40** can be used to control whether the signal transmitting and receiving elements **120** actually transmit the ultrasonic signals. With this feature, the high voltage control module **40** may also be used to assist in avoiding interference caused by adjacent at least two signal transmitting and receiving elements. For example, the high voltage control module **40** may detect whether the adjacent at least two signal transmitting and receiving elements **120** receive the startup signal at the same time or in a specific time. If yes, the high voltage control module **40** may output a disable signal (for example, "0") for the at least two signal transmitting and receiving elements **120**; that is, no high voltage is applied to the at least two signal transmitting and receiving elements **120**. If not, the high voltage control module **40** may output an enable signal (for example, "1") for the at least two signal transmitting and receiving elements **120**; that is, a high voltage is applied to the at least two signal transmitting and receiving elements **120**. However, the invention is not limited thereto. In one embodiment, the high voltage control module **40** is a hardware circuit; while the invention is not limited thereto.

(36) Next, the delay module **50** is described. In one embodiment, the delay module **50** may control the signal transmitting periods of the adjacent at least two signal transmitting and receiving elements **120** to have a time interval (ΔT) in advance, so that the adjacent two signal transmitting and receiving elements **120** will not perform signal transmitting and receiving during the same period regardless of the startup sequence mode.

(37) Next, the pulse modulation module **60** is described. In one embodiment, because the fault of the controller **16** caused by certain factors during the manufacturing process may result in the micro detection and treatment device **10** failing to output the correct power, the first power signal or the second power signal may be affected, resulting in a negative influence to the image-capturing or treatment quality. Therefore, the controller **16** may be configured with a pulse modulation module **60** for performing pulse width modulation on the signals inputted to the controller **16** or the signals outputted from the controller **16** so as to correct and enhance the signals, thereby allowing the micro detection and treatment device **10** to output the power of stable ultrasonic signal. However, the invention is not limited thereto.

(38) Next, the detailed features of the signal transmitting and receiving element array **12** are described. FIG. 3(A) schematically illustrates a signal transmitting and receiving element array according to an embodiment of the present invention. As shown in FIG. 3(A), the micro detection and treatment device **10** may have a cylindrical probe portion **13**, and the signal transmitting and receiving element array **12** surrounds the probe portion **13** by, for example, 360 degrees. In one example, the signal transmitting and receiving element array **12** may be provided with 64 signal transmitting and receiving elements **120**, which are equally arranged into an array and are divided into 4 rows, each row having 16 elements, each element being assigned with an address parameter (shown as brackets and numbers in the figure), but are not limited thereto. In other embodiments, the signal transmitting and receiving element array **12** may also be provided with other quantity of signal transmitting and receiving elements **120**, such as 16, 32, 128 or 256. In one embodiment, the signal transmitting and receiving element array **12** and the probe portion **13** may be provided with a

pivoting mechanism. Therefore, the signal transmitting and receiving element array **12** is rotatable relative to the probe portion **13**, but is not limited thereto. In one embodiment, the material of the signal transmitting and receiving element **120** is PZT (lead-zirconate-titanate), but is not limited thereto. In addition, in one embodiment, intervals may be provided between different rows, and thus each row may be regarded as a ring, but is not limited.

(39) FIG. 3(B) schematically illustrates a user operating device corresponding to the signal transmitting and receiving element array **12** according to an embodiment of the present invention. With reference to FIG. 3(B) and FIGS. 1(A) to 3(A), in one embodiment, the image (target object **500** and treatment portion **510**) obtained by the image-capturing procedure may be displayed on a display device **160**, and the address parameter of each signal transmitting and receiving element **120** may correspond to a lookup table. The user operating device **150** may be provided with a display area **52** for correspondingly displaying each address parameter according to the lookup table, and presenting each address parameter in a form of user-selectable key. Therefore, the user may directly select the desired signal transmitting and receiving element **120** on the user operating device **150** according to the treatment portion **510** of the target object **500** and, after the selection, uses a startup button **153** on the user operating device **150** to transmit a control command to the micro detection and treatment device **10**. In addition, in another embodiment, the display device **160** can be integrated into the user operating device **150**; that is, the image obtained by the image-capturing procedure can be displayed in the display area **52** at the same time with the address parameter of each signal transmitting and receiving element **120**. However, the present invention is not limited thereto.

(40) In one embodiment, when the micro detection and treatment device **10** transmits the “treatment signal (e.g. the second power signal)”, the display device **160** simultaneously displays the image-capturing of the treatment target, and the change of the treatment target can be presented. Thus, various treatment parameters (e.g. the treatment signal power, treatment period, etc.) can be adjusted in real time. In addition, since the display device **160** can display image-capturing in real time during the treatment process, the user or the micro detection and treatment device **10** can obtain the actual treatment area through the ratio conversion of the ultrasound image (e.g. correlation coefficient), and then determine whether to adjust the treatment position. In one embodiment, “simultaneously displaying the ultrasound images” can be achieved at least by: using the energy reflected from the treatment location to execute the piezoelectricity image-capturing on the same axis space of the treatment location.

(41) In one embodiment, since the display device **160** can display images in real time during the treatment process, the biological object image-capturing and treatment system **1** of the present invention can provide a function of monitoring the treatment process. In addition, since the monitoring function can be performed before or during the treatment process, the accuracy of the treatment can be improved.

(42) The user operating device **150** may have various implementations, for example, the user operating device **150** can be integrated with the micro detection and treatment device **10** or the user operating device **150** has a handle, but it is not limited thereto.

(43) The functions of the startup sequence setting module **30** and the startup module **35** of the present invention may be implemented by a digital logic circuit, which is described by way of an example in the following, but the invention is not limited to the example. FIG. 4 is a circuit diagram of the mode control module **30** and the startup module **35** according to an embodiment of the invention, and please refer to FIG. 1 to FIG. 3(B) at the same time.

(44) As shown in FIG. 4, the detailed structure of the mode control module **30** and the startup module **35** may include a signal generator **32**, a shift register unit **34**, and a latch data unit **36**. The shift register module **34** is coupled to the signal generator **32**, and includes 1-st to N-th startup sequence sub-circuits **341-34N**, wherein each of the startup sequence sub-circuits **341-34N** corresponds to a startup sequence mode. The latch data unit **36** outputs a latch signal, and the latch

signal is latched with the output of each of the startup sequence sub-circuits **341-34N**.

(45) In one embodiment, the signal generator **32** receives a first clock signal (Clk1) and a data signal (Data), and generates a second clock signal (hereinafter referred to as an enable signal (EN)) according to a first clock signal (Clk1) and the data signal (Data). The shift register module **34** receives the enable signal, and multiplexes and shifts the data signal (Data) according to the enable signal (EN), thereby generating N address signals (S1~SN), wherein each of the address signals (S1~SN) controls the output signal (i.e., startup signal) of one of the startup sequence sub-circuits **341-34N**, thereby controlling whether the startup sequence mode is ready to start. The latch data unit **36** performs an AND operation on the output signals (i.e., start signals) of the startup sequence sub-circuits **341-34N** according to the data signal (Data) and a third clock signal (Clk3). As a result, one of the startup sequence modes is actually started, and thus the signal transmitting and receiving elements **120** transmit or receive signals according to the startup sequence mode.

(46) In one embodiment, the first clock signal (clk1), the data signal (data), and the third clock signal (clk3) may be regarded as the control command transmitted by the user to the micro detection and treatment device **10** through the user operating device **150**; for example, a doctor assigns and selects signal transmitting and receiving elements **120** of specific address parameters for the portion to be treated.

(47) In one embodiment, the detailed structure may include at least one AND gate, wherein each AND gate is used to perform a logic operation on one of the address signals (S1~SN) and one of the startup signals (corresponding to one of the signal transmitting and receiving elements). In one embodiment, the quantity of the AND gates is equal to the product of the quantity of address signals (S1~SN) and the quantity of the startup signals.

(48) As a result, the signal transmitting and receiving elements **120** may perform signal transmitting and receiving according to a specific startup sequence mode, so as to prevent the adjacent signal transmitting and receiving elements **120** from performing signal transmitting and receiving in the same period.

(49) In one embodiment, each of the startup sequence sub-circuits **341~34N** includes a plurality of flip-flops. In one embodiment, each startup sequence sub-circuit includes 32 flip-flops. In one embodiment, in order to achieve the i-th startup sequence mode (i.e., the signal transmitting and receiving elements **120** are started in a manner of spacing i signal transmitting and receiving elements **120**), the key point is such that the output signal of the (i+1)-th flip of the i-th startup sequence sub-circuit **34i** has to be coupled to the input end of the 1-st flip-flop, while the subsequent flip flops are applied in a similar manner, so as to achieve the i-th startup sequence mode, where i is a positive integer less than or equal to N. As a result, the startup sequence sub-circuits **341~34N** may respectively implement the 1-st to N-th startup sequence modes.

(50) With the biological object image-capturing and treatment system and method of the present invention, it is able to not only integrate the image-capturing function and the treatment function into the same micro detection and treatment device **10**, but also greatly reduce the interference between adjacent signal channels through a startup sequence of special signal transmitting and receiving elements.

(51) Besides, the biological object image-capturing and treatment system **1** can be combined with an artificial intelligence (AI) model. FIG. 5 schematically illustrates the biological object image-capturing and treatment system **1** according to another embodiment of the present invention, and please refer to FIG. 1 to FIG. 4 at the same time. As shown in FIG. 5, the biological object image-capturing and treatment system **1** comprises the micro detection and treatment device **10**, the user operating device **150**, and an AI model **80**. The micro detection and treatment device **10** comprises a plurality of signal transmitting and receiving elements **120**, a transmitter **17**, a receiver **18** and a transmitting/receiving switch **19**.

(52) The biological object image-capturing and treatment system **1** according to this embodiment may be used for executing the object stimulation procedure, wherein the transmitter **17** may control

one of the signal transmitting and receiving elements **120** to transmit a stimulation signal to a particular target **550**. The particular target **550** receives the stimulation signal and reflects a response signal. The receiver **18** may receive the response signal through one of the signal transmitting and receiving elements **120**. The AI model **80** can perform machine learning.

(53) FIG. **6** is a flow chart illustrating the steps of the object stimulation procedure of the biological object image-capturing and treatment method according to an embodiment of the present invention, and please refer to FIG. **5** at the same time.

(54) As shown in FIG. **6**, first, step **S61** is executed, in which the micro detection and treatment device **10** receives an instruction from the user operating device **150**. Then, step **S62** is executed, in which the transmitter **170** selects one of the signal transmitting and receiving elements **120**, and controls the one of the signal transmitting and receiving elements **120** to continuously transmit the stimulation signal with different parameter values to the particular target **550**. Then, step **S63** is executed, in which the receiver **18** continuously receives the response signal corresponding to the stimulation signal with different parameter values through one of the signal transmitting and receiving elements **120**. Then, step **S64** is executed, in which the receiver **18** continuously inputs the response signal into the AI model **80**. Then, step **S65** is executed, the AI model **80** uses the response signals and stimulation results of the particular target **550** corresponding to the response signals to perform machine learning, and finds a parameter value of the stimulation signal most suitable for the particular target **550** according to the result of the machine learning. Then, step **S66** is executed, the AI model **80** outputs a parameter value regulation suggestion to the micro detection and treatment device **10** according to the parameter value of the stimulation signal most suitable for the particular target **550**. Then, step **S67** is executed, the transmitter **170** adjusts the parameter value of the stimulation signal transmitted from the one of the signal transmitting and receiving elements **120**. Then, step **S68** is executed, the processes of steps **S61** to **S67** are executed for other signal transmitting and receiving elements **120** until the parameter value of the stimulation signal transmitted from each of the signal transmitting and receiving elements **120** have been adjusted. Thus, the object stimulation procedure is completed. The sequence of the above steps can be changed as long as it is practicable.

(55) It is noted that, in this embodiment, after the adjustment of the parameter value of the stimulation signal transmitted from a signal transmitting and receiving elements **120** is completed, the adjustment of the parameter value of the stimulation signal transmitted from another signal transmitting and receiving element **120** is performed. However, in other embodiments, as long as the hardware resource of the systems **1** is sufficient, the adjustment of the parameter value of the stimulation signal transmitted from the plurality of signal transmitting and receiving elements **120** may also be performed simultaneously, for example, a plurality of AI models **80** can be used for different signal transmitting and receiving elements **120**.

(56) Besides, although the response signals are used as input variables of the AI model **80** according to this embodiment, the stimulation signal with different parameter values can also be used as the input variables of the AI model **80** in other embodiment. In addition, the one of the signal transmitting and receiving elements **120** transmitting the stimulation signal and the one of the signal transmitting and receiving elements **120** receiving the response signal may be same or different signal transmitting and receiving elements **120**.

(57) Next, the details of each element will be described.

(58) The signal transmitting and receiving elements **120** are applicable to the descriptions of the above embodiments. Besides, in the embodiment, each of the signal transmitting and receiving elements **120** corresponds to different particular target **550**, for example, to different target objects **500** (shown in FIG. **1**) or to different positions on the same target object **500**.

(59) In one embodiment, the transmitter **17** may be the controller **16** of the above embodiments, and has a signal generator **32** (as shown in FIG. **4**).

(60) In one embodiment, the receiver **18** may be a low voltage device with features of high

performance and low noise, but it is not limited thereto.

(61) In one embodiment, the transmitting/receiving switch **19** may be used to prevent the receiver **18** from receiving high voltage pulse from the controller **16**, and to allow the receiver **18** to receive the low voltage pulse of the response signal through the signal transmitting and receiving element **120**, thereby preventing the receiver **18** from being damaged by the high voltage pulse of the controller **16**. In one embodiment, the voltage of the response signal is smaller than or equal to ± 500 mV, but it is not limited thereto.

(62) In one embodiment, the AI model **80** may be an artificial neural network. The AI model **80** may continuously perform deep learning according to the response signal corresponding to the stimulation signal with different parameter values (e.g. different power) and the stimulation results of the particular target **550** corresponding to each response signal, and find the parameter value of the stimulation signal most suitable for the particular target **550** according to the result of the deep learning, so as to generate the parameter value adjustment suggestion. In one embodiment, the response signal may indicate the amount of energy absorbed by the particular target **550**, but it is not limited thereto.

(63) In one embodiment, the type of the parameter value may be power, frequency or amplitude, and it is not limited thereto. For the convenience of explanation, power is used as an example in the following.

(64) Furthermore, the object stimulation procedure may be performed inside a biological object or outside the biological object, as will be described separately below.

(65) When the object stimulation procedure is performed inside the biological object, the micro detection and treatment device **10** is inserted into the biological object, and each of the signal transmitting and receiving elements **120** transmits stimulation signals to different positions of the target object **500**, respectively, for example, each of the signal transmitting and receiving elements **120** corresponds to a position of the same organ, but it is not limited thereto.

(66) When the object stimulation procedure is performed outside the biological object, the micro detection and treatment device **10** is inserted outside the biological object, and each of the signal transmitting and receiving elements **120** transmits stimulation signals to different particular targets **550**, respectively, for example, one of the signal transmitting and receiving elements **120** can transmit the stimulation signal to an experimental sample, and another signal transmitting and receiving element **120** can transmit the stimulation signal to another experimental sample, but it is not limited thereto.

(67) Besides, the first power signal of the image-capturing procedure or the second power signal of the treatment procedure according to the aforementioned embodiments can be adjusted by using the biological object image-capturing and treatment system **1** of FIG. 5 and the biological object image-capturing and treatment method of FIG. 6, wherein the AI model **80** can find a parameter value of the first power signal or the second power signal that is most suitable for the biological object or different positions of the biological object.

(68) In this way, with the invention, it is able to automatically adjust power value of the stimulation signal transmitted by different signal transmitting and receiving elements **120** through the AI model **80**, so as to provide power value most suitable for the particular target **550**, thereby improving the stimulation effect.

(69) Although the present invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention as hereinafter claimed.

Claims

1. A biological object image-capturing and treatment system for performing an object stimulation procedure, comprising: a micro detection and treatment device including a plurality of signal

transmitting and receiving elements, wherein one of the signal transmitting and receiving elements continuously transmits a stimulation signal with different parameter values to a particular target, and one of the signal transmitting and receiving elements continuously receives response signal corresponding to the stimulation signal with different parameter values from the particular target; and an AI model using response signals corresponding to the stimulation signal with different parameter values from the particular target to perform machine learning, so as to output a parameter value adjustment suggestion to the micro detection and treatment device based on result of the machine learning, wherein the micro detection and treatment device adjusts the parameter value of the stimulation signal according to the parameter value adjustment suggestion, wherein a transmitter adjusts at least one of the parameter values of the stimulation signal transmitted from the one of the signal transmitting and receiving elements, wherein type of the at least one of the parameter values includes power, frequency and amplitude, wherein a diameter of each of the plurality of signal transmitting and receiving elements is equal to or smaller than 10000 micrometers, and the micro detection and treatment device is capable of being inserted into a human body; wherein the biological object image-capturing and treatment system is further for a treatment procedure, and the treatment procedure comprises a first treatment mode and a second treatment mode, the first treatment mode is set to transmit a power signal to different positions of a target object by using one of the plurality of signal transmitting and receiving elements, the second treatment mode is set to transmit the power signal to the different positions of the target object by using at a part of the plurality of signal transmitting and receiving elements; wherein the micro detection and treatment device further comprises a startup sequence setting module and a startup module, the startup sequence setting module controls the plurality of signal transmitting and receiving elements to perform one of a plurality of startup sequence modes for shifting transmitting timing and receiving timing of adjacent at least two elements of the plurality of signal transmitting and receiving elements, and the startup module generates a plurality of startup signals according to the startup sequence mode; wherein, a detailed structure of the startup sequence setting module and the startup module comprises a signal generator, a shift register unit, and a latch data unit, wherein the shift register unit is coupled to the signal generator and comprises a plurality of startup sequence sub-circuits, wherein each of the startup sequence sub-circuits corresponds to a startup sequence mode of the plurality of startup sequence modes, and the latch data unit outputs a latch signal for performing an AND operation on the output of each of the startup sequence sub-circuits.

2. The biological object image-capturing and treatment system of claim 1, wherein the AI model is a deep learning model, and the stimulation signal is an ultrasonic signal.
3. The biological object image-capturing and treatment system of claim 1, wherein, the object stimulation procedure is performed in an biological object, the particular target is a particular position of a target object of the biological object, and each of the signal transmitting and receiving elements transmits a stimulation signal to different positions of the target object.
4. The biological object image-capturing and treatment system of claim 1, wherein, the object stimulation procedure is performed outside an biological object, the particular target is a target object, and each of the signal transmitting and receiving elements transmits a stimulation signal to different target objects.
5. A biological object image-capturing and treatment method performed by a biological object image-capturing and treatment system for performing an object stimulation procedure, wherein the system comprises a micro detection and treatment device and an AI model, and the micro detection and treatment device comprises a plurality of signal transmitting and receiving elements, the method comprising the steps of: continuously transmitting a stimulation signal with different parameter values to a particular target by one of the signal transmitting and receiving elements; continuously receiving response signal corresponding to the stimulation signal with different parameter values from the particular target by one of the signal transmitting and receiving elements; using response signals corresponding to the stimulation signal with different parameter

values to perform machine learning by the AI model, and outputting a parameter value adjustment suggestion based on result of the machine learning to the micro detection and treatment device by the AI model; and adjusting at least one of the parameter values of the stimulation signal based on the parameter value adjustment suggestion by the micro detection and treatment device, wherein a transmitter adjusts the at least one of the parameter values of the stimulation signal transmitted from the one of the signal transmitting and receiving elements, wherein type of the at least one of the parameter values includes power, frequency and amplitude, wherein a diameter of each of the plurality of signal transmitting and receiving elements is equal to or smaller than 10000 micrometers, and the micro detection and treatment device is for being inserted into a human body; wherein the biological object image-capturing and treatment system is further for a treatment procedure, and the treatment procedure comprises a first treatment mode and a second treatment mode, the first treatment mode is set to transmit a power signal to different positions of a target object by using one of the plurality of signal transmitting and receiving elements, the second treatment mode is set to transmit the power signal to the different positions of the target object by using at a part of the plurality of signal transmitting and receiving elements; wherein the micro detection and treatment device further comprises a startup sequence setting module and a startup module, the startup sequence setting module controls the plurality of signal transmitting and receiving elements to perform one of a plurality of startup sequence modes for shifting transmitting timing and receiving timing of adjacent at least two elements of the plurality of signal transmitting and receiving elements, and the startup module generates a plurality of startup signals according to the startup sequence mode; wherein, a detailed structure of the startup sequence setting module and the startup module comprises a signal generator, a shift register unit, and a latch data unit, wherein the shift register unit is coupled to the signal generator and comprises a plurality of startup sequence sub-circuits, wherein each of the startup sequence sub-circuits corresponds to a startup sequence mode of the plurality of startup sequence modes, and the latch data unit outputs a latch signal for performing an AND operation on the output of each of the startup sequence sub-circuits.

6. The biological object image-capturing and treatment method of claim 5, wherein the AI model is a deep learning model, and the stimulation signal is an ultrasonic signal.

7. The biological object image-capturing and treatment method of claim 5, wherein, the object stimulation procedure is performed in an biological object, the particular target is a particular position of a target object of the biological object, and each of the signal transmitting and receiving elements transmits a stimulation signal to different positions of the target object.

8. The biological object image-capturing and treatment method of claim 5, wherein, the object stimulation procedure is performed outside a biological object, the particular target is a target object, and each of the signal transmitting and receiving elements transmits a stimulation signal to different target objects.
